

1. Administrative & Study Identification

Full Study Title: Clinical and Demographic Characteristics of Adult Patients with NEurofibromatosis in RUSSia **Short Title/ Acronym:** NEREUS **Protocol Number:** D134BR00007 **NCT Number:** NCT07088991 **Overall Status:** Active - Recruiting **Sponsor Name:** AstraZeneca **Investigational Product:** None (Observational/Non-Interventional; monitoring of routine care therapies such as MEK-inhibitors) **Phase of Development:** N/A (Observational) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective: To evaluate the clinical and demographic characteristics, as well as patient-reported outcomes (PROs), of adult patients with Neurofibromatosis Type 1 (NF1) and plexiform neurofibromas (PN) in a routine care setting in Russia. **Secondary Objectives:** To observe real-world treatment patterns (e.g., pathogenetic antitumor therapy), disease progression, PN-associated symptoms, and the impact of therapy modifications.

2.2 Background & Rationale To address the gap in real-world evidence regarding the clinical profile and treatment burden of adult NF1 patients with confirmed PN in Russia. By evaluating newly diagnosed patients or those established with PN who are naïve to MEK-inhibitor therapy, this study seeks to understand routine clinical management and long-term outcomes to improve future care pathways for this rare genetic disorder.

3. Study Design & Methodology

3.1 Design Framework

Study Type: Observational (Cohort) **Control Type:** Single-arm (Non-interventional) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration

Target \$N\$: 200 evaluable patients **Number of Sites:** Multi-center (approx. 12+ sites across the Russian Federation, including Arkhangelsk, Chelyabinsk, Krasnoyarsk, Moscow, Nizhny Novgorod, Novosibirsk, Saint Petersburg, Samara, Tomsk, Ufa, and Yekaterinburg) **Estimated Study Start:** 06-Jun-2025 **Estimated Primary Completion:** 31-Mar-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of inclusion. 2. Sex/Gender: All Genders. 3. Signed and dated written informed consent in accordance with ICH-GCP and local law prior to inclusion in the study. 4. Diagnosed with NF1 according to the international consensus criteria. 5. Confirmed Plexiform Neurofibroma (PN) by clinical assessment, ultrasound (USI), MRI, or biopsy, with existing PN-associated symptoms. 6. Newly diagnosed PN or established PN naïve to MEK-inhibitor therapy verified by medical records.

4.2 Exclusion Criteria 1. Current participation in any interventional clinical study (participation in other non-interventional studies is permitted). 2. Evidence of a malignant glioma, malignant peripheral nerve sheath tumor (MPNST), or other cancer requiring treatment with chemotherapy or radiation therapy. 3. Pregnant or breastfeeding individuals who are unable to undergo radiographic evaluations or MRI scans requested for research purposes, or other studies which might negatively impact the pregnancy. 4. Prior receipt of any MEK-inhibitor for PN therapy within 4 months before screening or initiation of such therapy prior to age 18 verified by medical records/histories.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Dosage/Frequency: N/A (Non-interventional study tracking routine standard of care) **Route of Administration:** N/A **Duration of Treatment:** Observational follow-up spanning the study timeframe (up to ~33 months).

5.2 Concomitant & Prohibited Therapy

Permitted: Standard of care therapies for NF1/PN as prescribed by the treating physician in routine practice. **Prohibited (for primary analysis):** Patients who modify their index pathogenetic therapy regimen (e.g., switching to a different pathogenetic antitumor therapy for PN) during the study will be excluded from the primary efficacy analysis population from the point of modification onward and will not continue study participation.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Verification of NF1/PN diagnosis Baseline Day 0 Clinical baseline characteristics, demographics (Age, Gender), Patient-Reported Outcomes (PROs) Interim Routine

Observational tracking of routine care visits, treatment modifications, and ongoing PROs | | **End of Study** | Mar 2028 | Final medical record extraction, final PRO evaluation, exit data closure |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Age (full years) at baseline; evaluation of clinical and patient-reported outcomes (PROs) in a routine care setting. **Measurement Method:** Abstraction from medical records/histories and completion of validated PRO questionnaires.

7.2 Secondary & Exploratory Endpoints Real-world treatment patterns and the incidence of pathogenetic therapy modifications. Symptomatic progression and long-term clinical evolution of Plexiform Neurofibromas.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collected from routine care medical records according to local real-world evidence and pharmacovigilance standards. **Serious Adverse Events (SAEs):** Reported per standard local regulatory timelines for observational studies. **Data Monitoring Committee (DMC):** N/A (Non-interventional design).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Primary Efficacy Analysis Population (excludes patients from the point they modify their index pathogenetic therapy). **Sample Size Justification:** $N=200$ powered for robust descriptive epidemiological profiling and qualitative real-world outcome assessments. **Handling of Missing Data:** As-observed descriptive statistics appropriate for observational cohort registries; missing data will not be imputed unless specified for specific PRO instruments.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (ICH-GCP) and local Russian federation laws. **Monitoring Plan:** Source data verification against patient medical records and electronic data capture (EDC) reviews. **Audit Trail:** Continuous audit trail utilizing secure EDC with locked data clarification forms. **Data Sharing:** AstraZeneca will provide access to anonymized patient-level data via the secure research environment vivli.org once the request is approved.

11. Study Identifiers & Governance

Primary ID: D134BR00007 **Registry Number:** NCT07088991 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** NEREUS **Official Regulatory Title:** Clinical and Demographic Characteristics of Adult Patients with NEurofibromatosis in RUSSia

12. Clinical Framework (Neurofibromatosis Specific)

Primary Condition: Neurofibromatosis Type 1 (NF1) **Diagnosis Threshold:** Confirmed Plexiform Neurofibroma (PN) with associated symptoms **Medical Methodology:** Real-World Routine Clinical Care **Optimization Target:** Disease stabilization and symptom management via index pathogenetic therapy

13. Study Procedures & Methodology

Management Pathway: Standard routine clinical pathway for NF1/PN **Education Protocol (HCP):** Protocol requirements for observational data extraction and PRO administration **Education Protocol (Patient):** Informed consent and PRO questionnaire guidance **Quality Audit Mechanism:** Electronic medical record cross-verification and validation of patient-reported entries

14. Observation & Follow-Up Schedule

Total Duration: ~33 months (June 2025 to March 2028) **Onsite Visit Cadence:** Aligned completely with routine standard of care visits **Remote Monitoring Frequency:** Intermittent remote data extraction **Checklist Review Interval:** According to scheduled routine clinical follow-ups

15. Sign-off & Approval

Role	Name	Signature	Date		:--	:--	:--	:--	
Principal Investigator	[Name]	____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]						
Statistician	[Name]	____	[DD-MMM-YYYY]						